Tutorial 1 – OLS Mechanics and Interpretation

Complete Theory Solutions and Python Lab Outputs

Introductory Econometrics – Telecom Paris

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1 Theory Exercises

1.1 Exercise 1 – Deriving OLS

Consider the simple regression model

$$y_i = \beta_0 + \beta_1 x_i + u_i, \qquad i = 1, \ldots, n.$$

(a) Sum of squared residuals and first-order conditions

For arbitrary candidate coefficients (b_0, b_1), the residual for observation i is

$$e_i(b_0, b_1) = y_i - b_0 - b_1 x_i.$$

OLS chooses $(\hat{\beta}_0, \hat{\beta}_1)$ to minimize

$$S(b_0, b_1) = \sum_{i=1}^{n} (y_i - b_0 - b_1 x_i)^2.$$

The first-order condition with respect to b_0 is

$$\frac{\partial S}{\partial b_0} = -2 \sum_{i=1}^{n} (y_i - b_0 - b_1 x_i) = 0,$$

so at the OLS solution,

$$\boxed{\sum_{i=1}^{n} \hat{u}_i = 0}.$$

The first-order condition with respect to b_1 is

$$\frac{\partial S}{\partial b_1} = -2 \sum_{i=1}^{n} x_i (y_i - b_0 - b_1 x_i) = 0,$$

so

$$\boxed{\sum_{i=1}^{n} x_i \hat{u}_i = 0}.$$

(b) OLS intercept and slope

From the first condition,

$$0 = \sum_i (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = n\bar{y} - n\hat{\beta}_0 - n\hat{\beta}_1 \bar{x},$$

therefore

$$\boxed{\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}}.$$

Substituting this into the second first-order condition gives

$$\sum_i (x_i - \bar{x})(y_i - \bar{y}) = \hat{\beta}_1 \sum_i (x_i - \bar{x})^2.$$

Hence

$$\boxed{\hat{\beta}_1 = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sum_i (x_i - \bar{x})^2}}.$$

The slope exists and is uniquely identified when

$$\boxed{S_{xx} = \sum_i (x_i - \bar{x})^2 > 0},$$

meaning that the sample contains variation in x; the x_i cannot all be equal.

(c) Sample analogues of the population moment conditions

The population conditions are

$$\mathbb{E}(u) = 0, \qquad \mathbb{E}(xu) = 0.$$

Their sample analogues replace expectations by sample averages and the unobserved population error by the OLS residual:

$$\frac{1}{n}\sum_i \hat{u}_i = 0, \qquad \frac{1}{n}\sum_i x_i\hat{u}_i = 0.$$

Multiplying by n gives exactly the two OLS first-order conditions derived above.

(d) Mechanical OLS identities

The first-order conditions directly imply

$$\boxed{\sum_i \hat{u}_i = 0}, \qquad \boxed{\sum_i x_i\hat{u}_i = 0}.$$

Moreover,

$$0 = \bar{\hat{u}} = \bar{y} - \hat{\beta}_0 - \hat{\beta}_1\bar{x},$$

so

$$\boxed{\bar{y} = \hat{\beta}_0 + \hat{\beta}_1\bar{x}}.$$

Thus the fitted OLS line passes through $(\bar{x}, \bar{y})$. These facts are *not* evidence that the population assumptions are correct: they hold mechanically because OLS chooses the coefficients to satisfy the sample first-order conditions.

1.2 Exercise 2 – Unbiasedness and Failure of Exogeneity

Let

$$S_{xx} = \sum_i (x_i - \bar{x})^2.$$

(a) Decomposition of the OLS slope

Start from the centered-sums formula:

$$\hat{\beta}_1 = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{S_{xx}}.$$

Since $\sum_i (x_i - \bar{x}) = 0$,

$$\sum_i (x_i - \bar{x})(y_i - \bar{y}) = \sum_i (x_i - \bar{x})y_i.$$

Substitute $y_i = \beta_0 + \beta_1 x_i + u_i$:

$$\begin{aligned}\hat{\beta}_1 &= \frac{\sum_i (x_i - \bar{x})(\beta_0 + \beta_1 x_i + u_i)}{S_{xx}} \\ &= \frac{\beta_0 \sum_i (x_i - \bar{x}) + \beta_1 \sum_i (x_i - \bar{x})x_i + \sum_i (x_i - \bar{x})u_i}{S_{xx}}.\end{aligned}$$

The first term is zero, and

$$\sum_i (x_i - \bar{x})x_i = \sum_i (x_i - \bar{x})^2 = S_{xx}.$$

Therefore

$$\boxed{\hat{\beta}_1 = \beta_1 + \frac{\sum_i (x_i - \bar{x})u_i}{S_{xx}}}.$$

(b) Conditional unbiasedness of the slope

Condition on the complete regressor sample $X = (x_1, \ldots, x_n)$. Then the x_i, $\bar{x}$, and S_{xx} are fixed. Hence

$$\mathbb{E}(\hat{\beta}_1 \mid X) = \beta_1 + \frac{\sum_i (x_i - \bar{x})\mathbb{E}(u_i \mid X)}{S_{xx}}.$$

Under the zero conditional mean assumption (SLR.4),

$$\mathbb{E}(u_i \mid X) = 0,$$

so

$$\boxed{\mathbb{E}(\hat{\beta}_1 \mid X) = \beta_1}.$$

It is precisely the zero conditional mean assumption that makes the numerator have conditional mean zero.

(c) Conditional unbiasedness of the intercept

Using $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$ and

$$\bar{y} = \beta_0 + \beta_1 \bar{x} + \bar{u},$$

we obtain

$$\hat{\beta}_0 - \beta_0 = \bar{u} - (\hat{\beta}_1 - \beta_1)\bar{x}.$$

Conditional on X,

$$\mathbb{E}(\bar{u} \mid X) = 0, \qquad \mathbb{E}(\hat{\beta}_1 - \beta_1 \mid X) = 0.$$

Therefore

$$\boxed{\mathbb{E}(\hat{\beta}_0 \mid X) = \beta_0}.$$

(d) Failure of exogeneity

Suppose instead that

$$\mathbb{E}(u_i \mid X) = \rho x_i.$$

Then

$$\begin{aligned}
\mathbb{E}(\hat{\beta}_1 \mid X) &= \beta_1 + \frac{\sum_i (x_i - \bar{x})\rho x_i}{S_{xx}} \\
&= \beta_1 + \rho \frac{\sum_i (x_i - \bar{x}) x_i}{S_{xx}} \\
&= \boxed{\beta_1 + \rho}.
\end{aligned}$$

If additionally $\mathbb{E}(x_i) = 0$, then $\mathbb{E}(u_i) = \rho \mathbb{E}(x_i) = 0$ and

$$\begin{aligned}
\mathrm{Cov}(x_i, u_i) &= \mathbb{E}(x_i u_i) - \mathbb{E}(x_i)\mathbb{E}(u_i) \\
&= \mathbb{E}[x_i \mathbb{E}(u_i \mid x_i)] \\
&= \rho \mathbb{E}(x_i^2) = \boxed{\rho \,\mathrm{Var}(x_i)}.
\end{aligned}$$

(e) Omitted ability in a wage regression

Suppose the true wage equation contains ability:

$$wage_i = \beta_0 + \beta_1 educ_i + \gamma ability_i + \varepsilon_i, \qquad \gamma > 0.$$

If ability is omitted, it becomes part of the regression error. If ability is also positively related to education, then high-education observations tend to have positive omitted components in the error. Consequently the covariance term in the slope decomposition tends to be positive, so the estimated education slope is biased upward relative to the partial effect β_1.

1.3 Exercise 3 – Unbiasedness and Precision

Let $Y_1, \ldots, Y_{10}$ be a random sample with $\mathbb{E}(Y_i) = \mu = 2$ and $\text{Var}(Y_i) = \sigma^2 = 1$. Consider

$$\tilde{\mu}_1 = Y_1, \qquad \tilde{\mu}_2 = \bar{Y}.$$

(a) Unbiasedness

For the first estimator,

$$\mathbb{E}(\tilde{\mu}_1) = \mathbb{E}(Y_1) = \mu = 2.$$

For the sample mean,

$$\mathbb{E}(\tilde{\mu}_2) = \mathbb{E}\left(\frac{1}{10}\sum_{i=1}^{10} Y_i\right) = \frac{1}{10}\sum_{i=1}^{10}\mathbb{E}(Y_i) = \mu = 2.$$

Thus

$$\boxed{\tilde{\mu}_1 \text{ and } \tilde{\mu}_2 \text{ are both unbiased.}}$$

(b) Variances and precision

$$\text{Var}(\tilde{\mu}_1) = \text{Var}(Y_1) = 1.$$

By independence,

$$\text{Var}(\tilde{\mu}_2) = \text{Var}(\bar{Y}) = \frac{1}{10^2}\sum_{i=1}^{10}\text{Var}(Y_i) = \frac{10}{100} = 0.1.$$

Therefore

$$\boxed{\text{Var}(\tilde{\mu}_1) = 1, \qquad \text{Var}(\tilde{\mu}_2) = 0.1.}$$

The sample mean has ten times smaller variance, so it is ten times more precise in the variance sense. Its standard deviation is $\sqrt{0.1} \approx 0.316$, compared with 1 for Y_1.

(c) Simulation of 20 samples from $N(2, 1)$

Using the supplied random seed, the 20 simulated values are:

Sample	Y_1	$\bar{Y}$
1	2.3047	1.6644
2	2.8794	2.2697
3	1.8151	2.1163
4	4.1416	2.1026
5	2.7433	2.3030
6	2.2891	1.9357
7	0.3171	2.0091
8	0.7243	1.8049
9	2.4568	1.8812
10	2.6654	1.4104
11	1.6218	1.7987
12	2.4344	1.9792
13	0.9765	2.0635
14	1.6102	2.2222
15	2.4144	1.6802
16	1.8414	1.7788
17	1.0349	2.0671
18	1.3452	1.8912
19	3.3060	2.3427
20	3.7237	2.0701

The simulation summaries are

$$\overline{Y_1} = 2.1323, \qquad \overline{\bar{Y}} = 1.9695,$$

with sample standard deviations 0.9907 and 0.2374, respectively. The observed ranges are

$$Y_1 : \ [0.3171, 4.1416], \qquad \bar{Y} : \ [1.4104, 2.3427].$$

Both empirical means are reasonably close to the population mean 2, but the sample-mean estimates are much more concentrated.

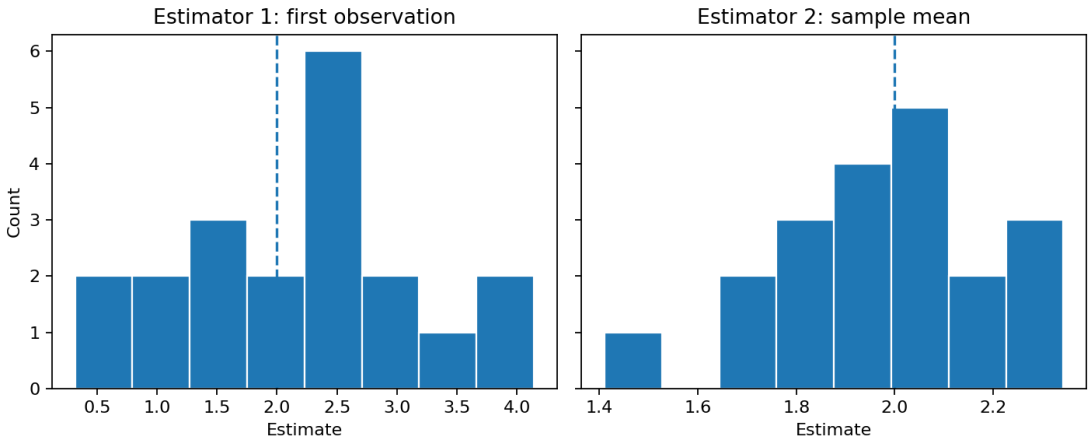


Figure 1: Exercise 3 simulation. The sample mean has a much tighter sampling distribution than the first observation.

(d) Unbiasedness does not imply accuracy in every sample

Unbiasedness is a statement about repeated sampling:

$$\mathbb{E}(\hat{\mu}) = \mu.$$

It does not imply that $\hat{\mu} = \mu$ in a particular sample. A single draw can be far from the true mean because the estimator has sampling variance. Comparing only average estimates therefore hides the key difference between the two procedures: both are centered correctly, but $\bar{Y}$ has much smaller variance and is therefore much more precise.

1.4 Exercise 4 – Estimating the Error Variance

Maintain SLR.1–SLR.5 and define

$$S_{xx} = \sum_i (x_i - \bar{x})^2, \qquad \text{SSR} = \sum_i \hat{u}_i^2.$$

(a) Decomposition of the residual sum of squares

Start from

$$\hat{u}_i = u_i - (\hat{\beta}_0 - \beta_0) - (\hat{\beta}_1 - \beta_1)x_i.$$

From Exercise 2,

$$d \equiv \hat{\beta}_1 - \beta_1 = \frac{\sum_i (x_i - \bar{x})u_i}{S_{xx}}.$$

Also,

$$\hat{\beta}_0 - \beta_0 = \bar{u} - d\bar{x}.$$

Therefore

$$\hat{u}_i = (u_i - \bar{u}) - d(x_i - \bar{x}).$$

Squaring and summing,

$$\begin{aligned} \text{SSR} &= \sum_i (u_i - \bar{u})^2 - 2d \sum_i (x_i - \bar{x})(u_i - \bar{u}) + d^2 \sum_i (x_i - \bar{x})^2 \\ &= \sum_i u_i^2 - n\bar{u}^2 - 2d \sum_i (x_i - \bar{x})u_i + d^2 S_{xx}. \end{aligned}$$

Let $A = \sum_i (x_i - \bar{x})u_i$, so $d = A/S_{xx}$. Then

$$-2dA + d^2 S_{xx} = -\frac{A^2}{S_{xx}}.$$

Hence

$$\boxed{\text{SSR} = \sum_i u_i^2 - n\bar{u}^2 - \frac{[\sum_i (x_i - \bar{x})u_i]^2}{S_{xx}}}.$$

(b) Conditional expectation of SSR

Under zero conditional mean and homoskedasticity,

$$\mathbb{E}(u_i \mid X) = 0, \qquad \text{Var}(u_i \mid X) = \sigma^2.$$

Thus

$$\mathbb{E}\left(\sum_i u_i^2 \mid X \right) = n\sigma^2.$$

Since $\mathbb{E}(\bar{u} \mid X) = 0$ and $\text{Var}(\bar{u} \mid X) = \sigma^2/n$,

$$\mathbb{E}(n\bar{u}^2 \mid X) = \sigma^2.$$

Finally, for

$$A = \sum_i (x_i - \bar{x})u_i,$$

we have $\mathbb{E}(A \mid X) = 0$ and

$$\mathrm{Var}(A \mid X) = \sigma^2 \sum_i (x_i - \bar{x})^2 = \sigma^2 S_{xx}.$$

Therefore

$$\mathbb{E}\left(\frac{A^2}{S_{xx}} \middle| X \right) = \sigma^2.$$

Putting the three terms together,

$$\boxed{\mathbb{E}(\mathrm{SSR} \mid X) = (n-2)\sigma^2}.$$

(c) Bias of $\tilde{\sigma}^2 = \mathrm{SSR}/n$

$$\begin{aligned} \mathbb{E}(\tilde{\sigma}^2 \mid X) &= \frac{1}{n}\mathbb{E}(\mathrm{SSR} \mid X) \\ &= \frac{n-2}{n}\sigma^2. \end{aligned}$$

Therefore

$$\boxed{\mathrm{Bias}(\tilde{\sigma}^2) = -\frac{2}{n}\sigma^2}.$$

The estimator SSR/n is downward biased.

(d) Unbiased estimator and degrees of freedom

Define

$$\hat{\sigma}^2 = \frac{\mathrm{SSR}}{n-2}.$$

Then

$$\mathbb{E}(\hat{\sigma}^2 \mid X) = \frac{1}{n-2}(n-2)\sigma^2 = \boxed{\sigma^2}.$$

Two degrees of freedom are lost because the residuals are computed after estimating two regression parameters, β_0 and β_1. Equivalently, the residual vector satisfies two independent sample restrictions:

$$\sum_i \hat{u}_i = 0, \qquad \sum_i x_i \hat{u}_i = 0.$$

1.5 Exercise 5 – Log-Level and Level-Log Specifications

Assume variables inside logarithms are strictly positive.

(a) Change in a logarithm

If $y_1 = y_0 + \Delta y$, then

$$\begin{aligned} \Delta \log y &= \log(y_1) - \log(y_0) \\ &= \log\left(\frac{y_0 + \Delta y}{y_0} \right) \\ &= \boxed{\log\left(1 + \frac{\Delta y}{y_0} \right)}. \end{aligned}$$

For a small $r = \Delta y/y_0$, the first-order Taylor approximation $\log(1+r) \approx r$ gives

$$\boxed{\Delta \log y \approx \frac{\Delta y}{y_0}}.$$

(b) Log-level model

For

$$\log(y) = \beta_0 + \beta_1 x + u,$$

a change Δx changes the fitted log outcome by

$$\Delta\widehat{\log y} = \beta_1 \Delta x.$$

Thus the approximate percentage change is

$$\boxed{100\beta_1 \Delta x\%}.$$

Between exponentiated fitted log values,

$$\frac{\exp(\hat{\ell}_1)}{\exp(\hat{\ell}_0)} = \exp(\beta_1 \Delta x),$$

so the exact percentage change is

$$\boxed{100\left[\exp(\beta_1 \Delta x) - 1\right]\%}.$$

(c) Level-log model

For

$$y = \beta_0 + \beta_1 \log(x) + u,$$

if x increases by $p\%$, then

$$x_1 = x_0\left(1 + \frac{p}{100}\right).$$

Therefore the exact change in fitted y is

$$\boxed{\Delta\hat{y} = \beta_1 \log\!\left(1 + \frac{p}{100}\right)}.$$

For a small percentage change,

$$\log\!\left(1 + \frac{p}{100}\right) \approx \frac{p}{100},$$

so

$$\boxed{\Delta\hat{y} \approx \beta_1 \frac{p}{100}}.$$

(d) Numerical applications

For the log-level model with $\hat{\beta}_1 = 0.08$:

- If $\Delta x = 1$, the approximation is 8%, while the exact change is

$$100(e^{0.08} - 1) = \boxed{8.33\%}.$$

- If $\Delta x = 4$, the approximation is 32%, while the exact change is

$$100(e^{0.32} - 1) = \boxed{37.71\%}.$$

For the level-log model with $\hat{\beta}_1 = 2.5$ and a 10% increase in x,

$$\Delta\hat{y}_{\text{approx}} = 2.5(0.10) = \boxed{0.25},$$

whereas

$$\Delta\hat{y}_{\text{exact}} = 2.5\log(1.10) = \boxed{0.2383}.$$

2 Python Lab 1 – Seeing Exogeneity and Bias by Simulation

The supplied simulation keeps the same x values and repeatedly estimates

$$y = 1 + 0.5x + u$$

under two data-generating processes:

$$\text{A: } u = \varepsilon, \qquad \text{B: } u = 0.4x + \varepsilon, \qquad \varepsilon \sim N(0, 4).$$

The completed OLS slope function is the centered-sums formula

$$\hat{\beta}_1 = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sum_i (x_i - \bar{x})^2}.$$

(a) Sampling distributions and numerical output

With $n = 200$ and 5,000 replications, the script produces:

Quantity	Case A	Case B
Mean simulated slope	0.502511	0.902511
Theoretical center	0.500000	0.900000

For Case A, the empirical variance of the slope is

$$0.0245403,$$

while the theoretical conditional variance is

$$\frac{\sigma^2}{S_{xx}} = 0.0258426.$$

The close agreement is what we expect from a large Monte Carlo experiment.

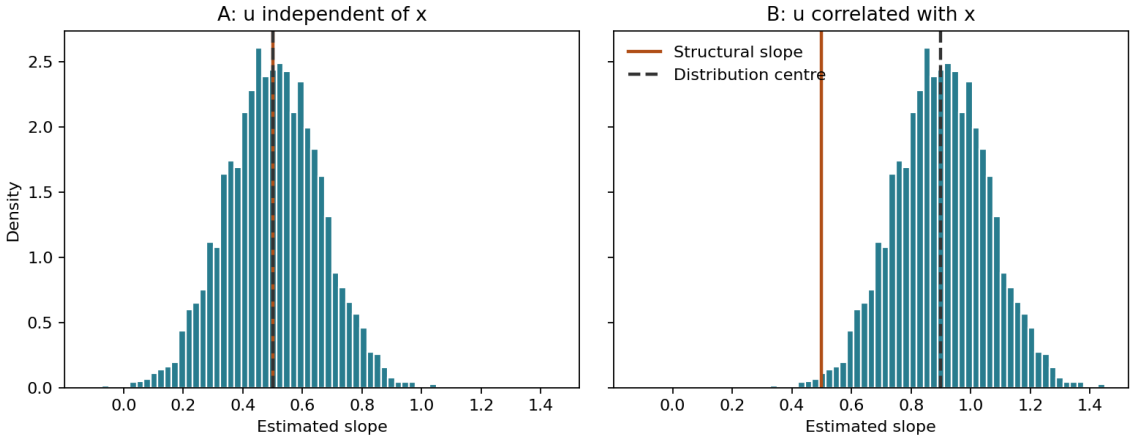


Figure 2: Sampling distributions of the OLS slope under exogeneity (A) and under correlation between x and the error (B).

(b) Interpretation

Under A, $\mathbb{E}(u \mid x) = 0$, so the OLS slope is unbiased and its sampling distribution is centered at the structural slope 0.5. A particular estimate can still be far from 0.5 because unbiasedness concerns the average across repeated samples, not every realization.

Under B,

$$u = 0.4x + \varepsilon \quad \Longrightarrow \quad \mathbb{E}(u \mid x) = 0.4x \neq 0,$$

so zero conditional mean fails. Exercise 2 predicts

$$\mathbb{E}(\hat{\beta}_1 \mid X) = 0.5 + 0.4 = 0.9,$$

which is exactly where the simulated distribution is centered. Increasing the number of Monte Carlo replications only estimates this biased sampling distribution more accurately; it does not move its center to 0.5.

3 Python Lab 2 – From Data to a Regression Interpretation

The data contain 526 workers and 24 variables from the 1976 Current Population Survey. The analysis uses hourly wage (`wage`) and years of education (`educ`) and estimates simple regressions with education as the only explanatory variable.

3.1 Step 1 – Inspect Before Estimating

(a) Data inspection

The data have 526 rows and 24 columns. For the two variables used here:

Statistic	Wage	Education
Count	526	526
Mean	5.8961	12.5627
Standard deviation	3.6931	2.7690
Minimum	0.5300	0
25th percentile	3.3300	12
Median	4.6500	12
75th percentile	6.8800	14
Maximum	24.9800	18

There are no missing values in either variable. One observation represents one worker; wage is measured in 1976 US dollars per hour and education in years. The education range is 0 to 18 years.

(b) Mean, median, positivity, and regression sample

The mean hourly wage is 5.8961, whereas the median is 4.6500. The mean exceeds the median because the wage distribution has a substantial right tail. The minimum wage is $0.53 > 0$, so all wages are strictly positive and the log wage can be computed. All 526 observations are therefore used in every fitted model.

(c) Required two descriptive sentences

One observation represents one worker; the sample contains 526 workers from the 1976 Current Population Survey. Wage is hourly earnings measured in 1976 US dollars, and education is measured in years.

3.2 Step 2 – Visualise and Form a Prediction

(a) Histograms and scatterplot

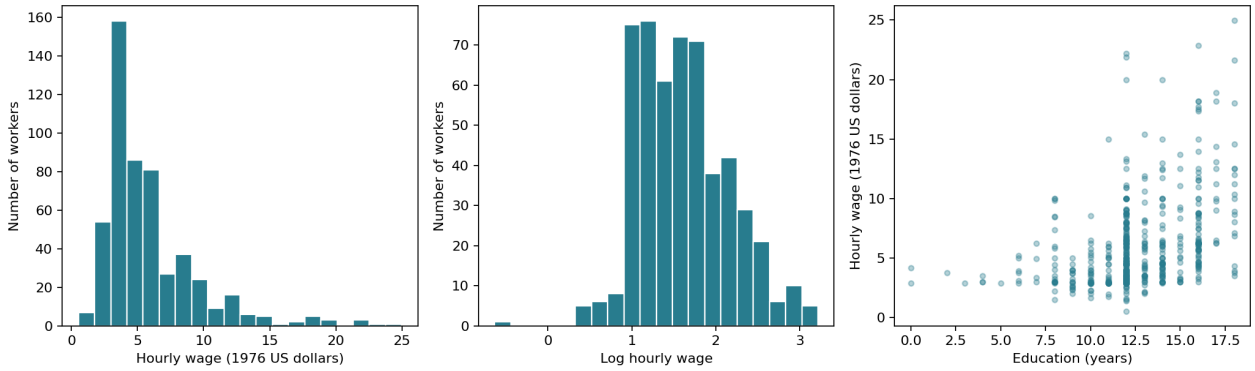


Figure 3: Distribution of wage, distribution of log wage, and the wage–education scatterplot.

The wage histogram is strongly right-skewed, with a small number of high-wage observations. Logging wage compresses the upper tail and produces a much less skewed distribution. The scatterplot suggests a positive relationship between education and wage, but the vertical spread at a given education level is large.

(b) Correlation and predicted slope sign

The sample correlation is

$$\boxed{\text{Corr}(wage, educ) = 0.4059}.$$

Thus the simple-regression slope should be positive. The scatterplot does not suggest accurate prediction of every worker's wage because workers with the same or similar education levels can have very different wages.

3.3 Step 3 – Fit and Read a Simple Regression

The estimated simple regression is

$$wage_i = a_0 + a_1 educ_i + v_i.$$

(a) Coefficients, sample size, and R^2

The software output gives

$$\hat{a}_0 = -0.904852, \qquad \hat{a}_1 = 0.541359, \qquad n = 526, \qquad R^2 = 0.164758.$$

Rounded as requested,

$$\boxed{\widehat{wage} = -0.905 + 0.541\, educ}, \qquad \boxed{n = 526, \quad R^2 = 0.165}.$$

(b) One and four additional years of education

One additional year of education is associated with

$$\boxed{\$0.541 \text{ more per hour}}.$$

Four additional years are associated with

$$4(0.541359) = \boxed{\$2.165 \text{ more per hour}}.$$

This is a simple regression containing only education, so it does *not* hold experience or any other worker characteristic fixed.

(c) Meaning of the negative intercept

The negative intercept means that the fitted line predicts approximately $-\$0.905$ at zero years of education. It does not imply that all fitted wages are negative, because fitted wage increases by about 0.541 for each additional year of education. The intercept is mainly the location where the fitted line crosses the vertical axis and is not economically useful here as a literal wage prediction.

(d) Centered-sums calculation and fitted line

Using the centered-sums formula gives

$$\hat{a}_1 = \frac{\sum_i (educ_i - \overline{educ})(wage_i - \overline{wage})}{\sum_i (educ_i - \overline{educ})^2} = 0.5413592547,$$

and

$$\hat{a}_0 = \overline{wage} - \hat{a}_1 \overline{educ} = -0.9048516120.$$

These match the software coefficients up to numerical precision.

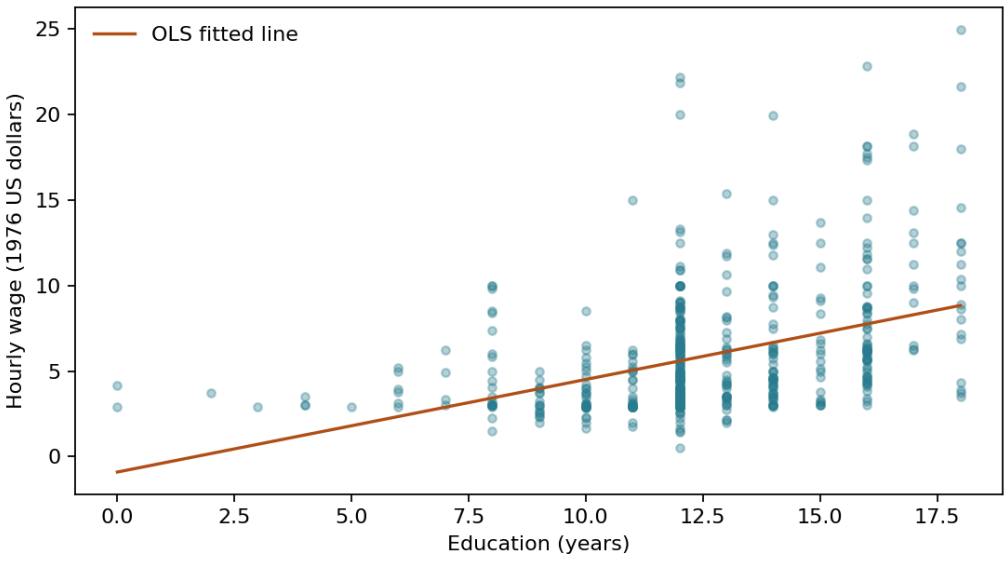


Figure 4: Observed hourly wages and the OLS fitted line.

(e) Required two interpretation sentences

One additional year of education is associated with an estimated \$0.541 *increase in hourly wage. Education is the only included regressor, so no other worker characteristic is held fixed and the estimated relationship is an association, not a causal effect.*

3.4 Step 4 – Fitted Values and Residuals

(a) Fitted wages at 12 and 16 years of education

The fitted values are

$$\widehat{wage}(12) = \boxed{5.591459}, \qquad \widehat{wage}(16) = \boxed{7.756896}.$$

Their difference is

$$7.756896 - 5.591459 = \boxed{2.165437},$$

which is exactly four times the estimated slope. These are fitted values from the regression line, not the known wage of every worker with 12 or 16 years of education.

(b) First residual

For the first observation, $educ_1 = 11$ and $wage_1 = 3.10$. The fitted wage is

$$\widehat{wage}_1 = 5.050100,$$

so

$$\hat{u}_1 = wage_1 - \widehat{wage}_1 = 3.10 - 5.050100 = \boxed{-1.950100}.$$

The residual is negative because this worker's actual wage lies below the fitted regression line. We can calculate the residual because the fitted coefficients are observed after estimation; we cannot calculate the population error u_i because the true population regression function and its true coefficients are unknown.

(c) OLS checks

Numerically,

$$\sum_i \hat{u}_i = -4.01 \times 10^{-12} \approx 0.$$

Also,

$$\widehat{wage}(\overline{educ}) = 5.896102675, \qquad \overline{wage} = 5.896102675.$$

Thus the residuals sum to zero up to floating-point error and the fitted line passes through $(\overline{educ}, \overline{wage})$. These facts do not validate exogeneity; they are mechanical consequences of OLS with an intercept.

(d) Residual plot

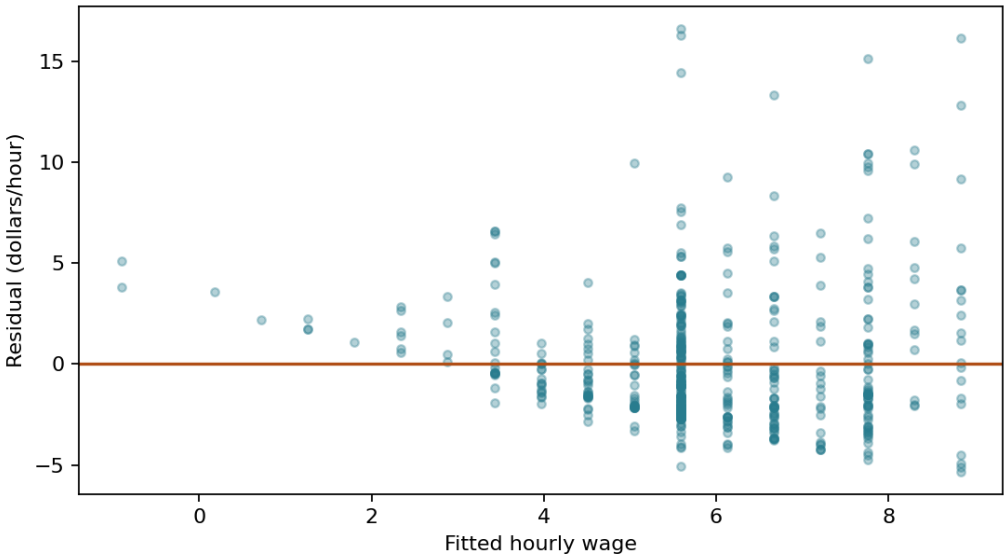


Figure 5: OLS residuals plotted against fitted hourly wages.

The residual spread appears to become wider at larger fitted wages, especially on the positive side. This is a descriptive visual pattern that may suggest non-constant variance, but the plot is not a formal test and it does not prove a causal problem.

(e) Required fitted-value sentence

At 12 years of education the regression predicts an hourly wage of about \$5.59*, which is a fitted value from the sample regression and not the actual wage of every worker with 12 years of education.*

3.5 Step 5 – Compare Specifications and Units

The three fitted models are

$$\text{A: } wage_i = a_0 + a_1 educ_i + v_i,$$

$$\text{B: } \log(wage_i) = b_0 + b_1 educ_i + e_i,$$

$$\text{C: } wage_cents_i = c_0 + c_1 educ_i + w_i, \qquad wage_cents_i = 100 wage_i.$$

(a) Model comparison table

Model	Outcome	Intercept	Slope	R^2	Slope units	n
A	wage	-0.904852	0.541359	0.164758	dollars/hour/year	526
B	log wage	0.583773	0.082744	0.185806	log points/year	526
C	wage cents	-90.485161	54.135925	0.164758	cents/hour/year	526

All three models use $n = 526$. The slopes from A and B cannot be compared as dollar amounts because they refer to different outcome units: Model A uses wage levels in dollars per hour, whereas Model B uses log wage.

(b) Approximate and exact percentage comparison in Model B

For one additional year of education,

$$100\hat{b}_1 = 100(0.0827444) = \boxed{8.274\%}.$$

The exact percentage difference between exponentiated fitted log wages is

$$100\left(e^{\hat{b}_1} - 1\right) = \boxed{8.626\%}.$$

Applying the same exact percentage to the conditional arithmetic mean $\mathbb{E}(wage \mid educ)$ requires the retransformation factor

$$\mathbb{E}(e^e \mid educ)$$

to be constant in education (for example, this follows if the distribution of the log error is independent of education). Without such a restriction, exponentiating a fitted log value does not by itself give the conditional arithmetic mean wage.

(c) Exponentiated fitted log wages

At 12 years of education,

$$\widehat{\log(wage)} = 1.576705, \qquad e^{1.576705} = \boxed{4.838985}.$$

At 16 years,

$$\widehat{\log(wage)} = 1.907683, \qquad e^{1.907683} = \boxed{6.737457}.$$

These are *exponentiated fitted log wages*. They should not automatically be called estimates of mean wages because generally

$$\mathbb{E}(wage \mid educ) = e^{\beta_0+\beta_1 educ}\mathbb{E}(e^e \mid educ),$$

and the final expectation need not equal one.

(d) Comparing the two R^2 values

Model B has $R^2 = 0.185806$, larger than Model A's 0.164758, but this does not establish that B predicts wage in dollars better. The two R^2 statistics use different dependent variables and therefore different total sums of squares:

$$\sum_i (wage_i - \overline{wage})^2 \quad \text{versus} \quad \sum_i (\log wage_i - \overline{\log wage})^2.$$

They measure fractions of variation on different scales and are not directly comparable for dollar prediction performance.

(e) Changing dollars to cents

Since $wage_cents = 100\, wage$, we expect the intercept, slope, fitted values, and residuals to be multiplied by 100, while R^2 remains unchanged. The estimates confirm this:

$$-90.485161 = 100(-0.904852), \qquad 54.135925 = 100(0.541359),$$

and the numerical checks for coefficients, residuals, and R^2 are all `True`. Expressing wages in cents changes only the units and provides no new economic information.

(f) Required three sentences

In Model A, one additional year of education is associated with about \$0.541 *more hourly wage. In Model B, one additional year is associated with approximately* 8.274% *higher wage and with an exact* 8.626% *difference between exponentiated fitted log wages. Neither estimate is automatically causal because education is the only included regressor and omitted factors such as ability can be related to both education and wages.*

4 Key Output Files Generated by the Completed Script

The completed script generates the following files in `code/output/`:

- `exercise3_sampling_distributions.png`
- `exercise3_comparison.csv`
- `lab1_sampling_distribution.png`
- `lab2_distributions_and_scatter.png`
- `lab2_fitted_line.png`
- `lab2_residuals.png`
- `lab2_predictions.csv`
- `lab2_specifications.csv`
- `console_output.txt`

A Complete Python Console Output

```
       first_observation  sample_mean
count          20.000000    20.000000
mean            2.132268     1.969549
std             0.990718     0.237418
min             0.317130     1.410378
25%             1.543946     1.803368
50%             2.296918     1.994146
75%             2.684852     2.106040
max             4.141648     2.342679
Ranges:
 first_observation: 0.31713022838419525 4.141647600870462
 sample_mean: 1.4103784926539185 2.342679229124143
Mean slope when u is independent of x: 0.5025110410714281
Target under exogeneity: 0.5
Mean slope when u is correlated with x: 0.9025110410714282
Target when E(u|x) = rho*x: 0.9
Empirical variance under exogeneity: 0.024540283131347467
Theoretical variance conditional on x: 0.025842575218030565
   wage  educ  exper  tenure  ...  servocc     lwage  expersq  tenursq
0  3.10    11      2       0  ...        0  1.131402        4        0
1  3.24    12     22       2  ...        1  1.175573      484        4
2  3.00    11      2       0  ...        0  1.098612        4        0
3  6.00     8     44      28  ...        0  1.791759     1936      784
4  5.30    12      7       2  ...        0  1.667707       49        4

[5 rows x 24 columns]
Rows and columns: (526, 24)
```

```
             wage        educ
count  526.000000  526.000000
mean     5.896103   12.562738
std      3.693086    2.769022
min      0.530000    0.000000
25%      3.330000   12.000000
50%      4.650000   12.000000
75%      6.880000   14.000000
max     24.980000   18.000000
Missing values:
 wage    0
educ    0
dtype: int64
Used / excluded: 526 0
Median wage: 4.650000095367432
Education range: (np.int64(0), np.int64(18))
Wage-education correlation: 0.40590332715075284
Levels coefficients:
 Intercept   -0.904852
educ         0.541359
dtype: float64
n and R squared: 526.0 0.1647575109920515
Centred-sums coefficients: -0.9048516119572119 0.5413592546651746
Software coefficients:
 Intercept   -0.904852
educ         0.541359
dtype: float64
   educ  fitted_wage
0    12     5.591459
1    16     7.756896
   educ  wage  fitted_wage  residual
0    11  3.10     5.050100 -1.950100
1    12  3.24     5.591459 -2.351459
2    11  3.00     5.050100 -2.050100
3     8  6.00     3.426022  2.573978
4    12  5.30     5.591459 -0.291459
Residual sum: -4.007461029686965e-12
Fitted at mean education: 5.896102674787043
Mean wage: 5.896102674787035
        model  intercept     slope                    slope_units   n       R2
      A: wage  -0.904852  0.541359 dollars/hour/year of education 526 0.164758
  B: log wage   0.583773  0.082744   log points/year of education 526 0.185806
C: wage cents -90.485161 54.135925   cents/hour/year of education 526 0.164758
   educ  fitted_wage  fitted_log_wage  exp_fitted_log_wage
0    12     5.591459         1.576705             4.838985
1    16     7.756896         1.907683             6.737457
Approximate percentage comparison: 8.274436586375078
Exact exponentiated-log comparison: 8.626408687059417
Cents checks: True True True
```

B Completed Python Script

```
"""Tutorial 1: OLS mechanics, simulation, and wage interpretation.

Student version. Keep the supplied folder structure and complete the items
marked TODO. The data and output paths are detected automatically.
The supplied sections continue running when a TODO is unfinished, so one
unfinished calculation does not prevent you from reaching the next section.

Requirements: numpy, pandas, matplotlib, statsmodels.
"""

# %% 0. Paths and imports - keep the supplied folder structure unchanged
```

```
from pathlib import Path

HERE = Path(__file__).resolve().parent
PATH_DATA = HERE.parent / "data" / "wage1.csv"
PATH_OUTPUT = HERE / "output"

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import statsmodels.formula.api as smf


def check_paths() -> None:
    """Check the supplied folder structure and create the output folder."""
    if not PATH_DATA.is_file():
        raise FileNotFoundError(
            f"Data file not found: {PATH_DATA}\n"
            "Extract the complete Tutorial 1 ZIP and keep the code and data "
            "folders together."
        )
    PATH_OUTPUT.mkdir(parents=True, exist_ok=True)


check_paths()
rng = np.random.default_rng(42)


def report_todo(label: str) -> None:
    print(f"TODO not completed: {label}")


# %% Exercise 3 simulation
exercise3_rng = np.random.default_rng(42)
samples = exercise3_rng.normal(loc=2, scale=1, size=(20, 10))
# TODO F1: extract the first observation and mean from each row.
first_observations = samples[:, 0]
sample_means = samples.mean(axis=1)
if first_observations is None or sample_means is None:
    report_todo("F1 Exercise 3 simulated estimators")
else:
    comparison = pd.DataFrame(
        {"first_observation": first_observations, "sample_mean": sample_means}
    )
    print(comparison.describe())
    comparison.to_csv(PATH_OUTPUT / "exercise3_comparison.csv", index=False)
    print("Ranges:")
    print(" first_observation:", first_observations.min(), first_observations.max())
    print(" sample_mean:", sample_means.min(), sample_means.max())
    fig, axes = plt.subplots(1, 2, figsize=(9, 3.8), sharey=True)
    axes[0].hist(first_observations, bins=8, edgecolor="white")
    axes[0].axvline(2, linewidth=1.5, linestyle="--")
    axes[0].set(title="Estimator 1: first observation", xlabel="Estimate", ylabel="Count")
    axes[1].hist(sample_means, bins=8, edgecolor="white")
    axes[1].axvline(2, linewidth=1.5, linestyle="--")
    axes[1].set(title="Estimator 2: sample mean", xlabel="Estimate")
    fig.tight_layout()
    fig.savefig(PATH_OUTPUT / "exercise3_sampling_distributions.png", dpi=160)
    plt.show()
    plt.close(fig)


# %% Lab 1 - seeing exogeneity and bias by simulation
def ols_slope(x: np.ndarray, y: np.ndarray):
    """Return the OLS slope for a regression of y on a constant and x."""
    # TODO L1.1: replace None with the centred-sums formula.
```

```
    # Hint: xd = x - x.mean(); yd = y - y.mean()
    xd = x - x.mean()
    yd = y - y.mean()
    return np.sum(xd * yd) / np.sum(xd ** 2)


n, n_rep = 200, 5000
beta1_true, rho, sigma = 0.5, 0.4, 2.0
x = rng.normal(size=n)  # keep the same x values in every replication
test_y = 1 + beta1_true * x + rng.normal(scale=sigma, size=n)

if ols_slope(x, test_y) is None:
    report_todo("L1.1 OLS slope function")
else:
    b1_exogenous = np.empty(n_rep)
    b1_correlated = np.empty(n_rep)
    for r in range(n_rep):
        epsilon = rng.normal(scale=sigma, size=n)
        y_exogenous = 1 + beta1_true * x + epsilon
        y_correlated = 1 + beta1_true * x + rho * x + epsilon
        b1_exogenous[r] = ols_slope(x, y_exogenous)
        b1_correlated[r] = ols_slope(x, y_correlated)

    sst_x = ((x - x.mean()) ** 2).sum()
    print("Mean slope when u is independent of x:", b1_exogenous.mean())
    print("Target under exogeneity:", beta1_true)
    print("Mean slope when u is correlated with x:", b1_correlated.mean())
    print("Target when E(u|x) = rho*x:", beta1_true + rho)
    print("Empirical variance under exogeneity:", b1_exogenous.var())
    print("Theoretical variance conditional on x:", sigma**2 / sst_x)

    fig, axes = plt.subplots(1, 2, figsize=(10, 4), sharex=True, sharey=True)
    for ax, estimates, target, title in [
        (axes[0], b1_exogenous, beta1_true, "A: u independent of x"),
        (axes[1], b1_correlated, beta1_true + rho, "B: u correlated with x"),
    ]:
        ax.hist(estimates, bins=50, density=True, color="#287c8e", edgecolor="white")
        ax.axvline(beta1_true, color="#b04e16", linewidth=2, label="Structural slope")
        ax.axvline(target, color="#333333", linewidth=2, linestyle="--",
                   label="Distribution centre")
        ax.set(xlabel="Estimated slope", title=title)
    axes[0].set_ylabel("Density")
    axes[1].legend(frameon=False)
    fig.tight_layout()
    fig.savefig(PATH_OUTPUT / "lab1_sampling_distribution.png", dpi=160)
    plt.show()
    plt.close(fig)

# WRITE L1.2: Explain why one estimate can be far from 0.5 under A and why the
# distribution under B is centred near 0.9. Identify the exogeneity condition
# that holds under A and fails under B. Would more replications remove the bias?


# %% Lab 2, Step 1 - inspect before estimating
raw = pd.read_csv(PATH_DATA)
variables = ["wage", "educ"]
print(raw.head())
print("Rows and columns:", raw.shape)
print(raw[variables].describe())
print("Missing values:\n", raw[variables].isna().sum())

keep = raw[["wage", "educ"]].notna().all(axis=1) & (raw["wage"] > 0)
df = raw.loc[keep].copy()
print("Used / excluded:", len(df), len(raw) - len(df))
df["log_wage"] = np.log(df["wage"])
```

```
df["wage_cents"] = 100 * df["wage"]

# TODO L2.1: calculate the median wage and the minimum and maximum education.
wage_median = df["wage"].median()
education_range = (df["educ"].min(), df["educ"].max())
if wage_median is None or education_range is None:
    report_todo("L2.1 median wage and education range")
else:
    print("Median wage:", wage_median)
    print("Education range:", education_range)

# WRITE L2.2: Define the sample, observation unit, wage units, education units,
# and the number of observations used in the regressions.


# %% Lab 2, Step 2 - visualise and form a prediction
fig, axes = plt.subplots(1, 3, figsize=(13, 4))
axes[0].hist(df.wage, bins=20, color="#287c8e", edgecolor="white")
axes[0].set(xlabel="Hourly wage (1976 US dollars)", ylabel="Number of workers")
axes[1].hist(df.log_wage, bins=20, color="#287c8e", edgecolor="white")
axes[1].set(xlabel="Log hourly wage", ylabel="Number of workers")
axes[2].scatter(df.educ, df.wage, alpha=0.35, s=14, color="#287c8e")
axes[2].set(xlabel="Education (years)", ylabel="Hourly wage (1976 US dollars)")
fig.tight_layout()
fig.savefig(PATH_OUTPUT / "lab2_distributions_and_scatter.png", dpi=160)
plt.show()
plt.close(fig)

# TODO L2.3: calculate the sample correlation between wage and education.
wage_education_correlation = df["wage"].corr(df["educ"])
if wage_education_correlation is None:
    report_todo("L2.3 wage-education correlation")
else:
    print("Wage-education correlation:", wage_education_correlation)

# WRITE L2.4: Predict the sign of the slope and explain why the scatterplot
# does not suggest accurate prediction of every worker's wage.


# %% Lab 2, Step 3 - fit and read a simple regression
levels = smf.ols("wage ~ educ", data=df).fit()
print("Levels coefficients:\n", levels.params)
print("n and R squared:", levels.nobs, levels.rsquared)

# TODO L2.5: reproduce the slope and intercept with centred sums.
xd = df["educ"] - df["educ"].mean()
yd = df["wage"] - df["wage"].mean()
centred_slope = (xd * yd).sum() / (xd ** 2).sum()
centred_intercept = df["wage"].mean() - centred_slope * df["educ"].mean()
if centred_slope is None or centred_intercept is None:
    report_todo("L2.5 centred-sums slope and intercept")
else:
    print("Centred-sums coefficients:", centred_intercept, centred_slope)
    print("Software coefficients:\n", levels.params)

grid = pd.DataFrame({"educ": np.linspace(df.educ.min(), df.educ.max(), 100)})
fig, ax = plt.subplots(figsize=(7, 4))
ax.scatter(df.educ, df.wage, alpha=0.35, s=15, color="#287c8e")
ax.plot(grid.educ, levels.predict(grid), color="#b04e16", label="OLS fitted line")
ax.set(xlabel="Education (years)", ylabel="Hourly wage (1976 US dollars)")
ax.legend(frameon=False)
fig.tight_layout()
fig.savefig(PATH_OUTPUT / "lab2_fitted_line.png", dpi=160)
plt.show()
plt.close(fig)
```

```
# WRITE L2.6: Report the fitted equation, interpret one and four additional
# years of education, state what is held fixed, and discuss the intercept.


# %% Lab 2, Step 4 - fitted values and residuals
workers = pd.DataFrame({"educ": [12, 16]})
workers["fitted_wage"] = levels.predict(workers)
print(workers)

df["fitted_wage"] = levels.fittedvalues
df["residual"] = levels.resid
print(df[["educ", "wage", "fitted_wage", "residual"]].head())

# TODO L2.7: calculate the two mechanical OLS checks.
residual_sum = df["residual"].sum()
fitted_at_mean_education = levels.params["Intercept"] + levels.params["educ"] * df["educ"].mean()
if residual_sum is None or fitted_at_mean_education is None:
    report_todo("L2.7 residual sum and fitted value at mean education")
else:
    print("Residual sum:", residual_sum)
    print("Fitted at mean education:", fitted_at_mean_education)
    print("Mean wage:", df.wage.mean())

fig, ax = plt.subplots(figsize=(7, 4))
ax.scatter(levels.fittedvalues, levels.resid, alpha=0.35, s=15, color="#287c8e")
ax.axhline(0, color="#b04e16")
ax.set(xlabel="Fitted hourly wage", ylabel="Residual (dollars/hour)")
fig.tight_layout()
fig.savefig(PATH_OUTPUT / "lab2_residuals.png", dpi=160)
plt.show()
plt.close(fig)

# WRITE L2.8: Calculate and interpret the first residual. Explain why the two
# OLS identities do not validate exogeneity. Report one fitted value while
# distinguishing it from an individual outcome.


# %% Lab 2, Step 5 - compare specifications and units
logs = smf.ols("log_wage ~ educ", data=df).fit()
cents = smf.ols("wage_cents ~ educ", data=df).fit()

specifications = pd.DataFrame(
    [
        {"model": label, "intercept": fit.params["Intercept"],
         "slope": fit.params["educ"], "slope_units": units,
         "n": int(fit.nobs), "R2": fit.rsquared}
        for label, fit, units in [
            ("A: wage", levels, "dollars/hour/year of education"),
            ("B: log wage", logs, "log points/year of education"),
            ("C: wage cents", cents, "cents/hour/year of education"),
        ]
    ]
)
print(specifications.to_string(index=False))
specifications.to_csv(PATH_OUTPUT / "lab2_specifications.csv", index=False)

workers["fitted_log_wage"] = logs.predict(workers)
workers["exp_fitted_log_wage"] = np.exp(workers["fitted_log_wage"])
print(workers)
workers.to_csv(PATH_OUTPUT / "lab2_predictions.csv", index=False)

# TODO L2.9: calculate the approximate and exact exponentiated-log percentage
# comparisons for one additional year of education.
approximate_percentage = 100 * logs.params["educ"]
```

```
exact_percentage = 100 * (np.exp(logs.params["educ"]) - 1)
if approximate_percentage is None or exact_percentage is None:
    report_todo("L2.9 approximate and exact percentage comparisons")
else:
    print("Approximate percentage comparison:", approximate_percentage)
    print("Exact exponentiated-log comparison:", exact_percentage)

# TODO L2.10: verify how the cents coefficients, residuals, and R squared
# relate to the dollar model. Store True or False in each item.
cents_coefficients_check = np.allclose(cents.params.values, 100 * levels.params.values)
cents_residuals_check = np.allclose(cents.resid.values, 100 * levels.resid.values)
cents_r2_check = np.isclose(cents.rsquared, levels.rsquared)
print("Cents checks:", cents_coefficients_check, cents_residuals_check, cents_r2_check)

# WRITE L2.11: Compare the dollar and percentage associations. Explain why the
# two R squared values are not directly comparable, why exponentiated fitted
# log wages are not automatically conditional mean wages, and why neither
# specification by itself identifies a causal return to education.
```